Prompt input/output reference — inputs, filled prompt, and output per call

Every LLM call in the current design, in pipeline order. For each: an **Inputs** list (field → what it is), the **filled prompt** the model receives (template variables substituted with data), and the **output** (structured-output schema + example). Generation and selection prompts read the ticket's **raw text** (`{raw text}` below = the full consolidated ticket content, ~24k tokens).

---

1. Condense (LLM ×1, per ticket)

Inputs:

- `ticketId` → the ticket id, for traceability.
- `consolidatedText` → the full raw text (description + extracted attachments, ~24k tokens).

Filled prompt (user message):

```
Ticket ID: IDMT-19761

Source material:
{raw text}
```

Output — `SummaryFields`:

```
{
  "generatedSummary": "Sales Operations needs a real-time CPQ integration to cut the enterprise quote cycle from five days to same-day, with automated discount approval and a handoff to order fulfilment.",
  "businessProblem": "Manual quoting delays enterprise deal closures by 3-5 days.",
  "businessCapability": "Automated real-time quote generation for enterprise accounts.",
  "keyTerms": ["CPQ", "quoting", "enterprise", "Salesforce", "Deal Desk"],
  "stakeholders": ["Sales Ops", "IT", "Finance"],
  "systemsAndProducts": ["Salesforce CPQ", "Oracle ERP", "Deal Desk Portal"]
}
```

*The summary fields are used for **retrieval** (embedding query) and routing — not fed into the generation prompts below.*

---

2. Value Stream Selection (LLM ×1, per ticket)

Inputs:

- `content` → the new ticket's **raw text** (what the model reads to decide). *(The summary is used separately as the embedding query to fetch the 6 past tickets — it is not in this prompt.)*

- `requestedCount` → how many Value Streams to return (exact; default 10).
- `historicEvidence` → the **6 similar past tickets**, each shown as its summary + the Value Streams it was tagged with (precedent).
- `candidateBlocks` → **all 50 governed Value Streams**, one compact block each.

Filled prompt (user message):

```
{raw text}

REQUESTED VALUE STREAM COUNT (exact):
10

SIMILAR PAST TICKETS (evidence - the value streams these were tagged with):
- IDMT-17432: CPQ enhancements for the large Deal Desk; automated pricing configuration for enterprise tiers.
  -> tagged value streams: Configure, Price and Quote (VSR-0042), Order Management (VSR-0017)
- IDMT-16890: Order fulfilment integration with CPQ; quote-to-order handoff latency reduction.
  -> tagged value streams: Order Management (VSR-0017)

CANDIDATE VALUE STREAMS:

Candidate: Configure, Price and Quote
entity_id: VSR-0042
description: Initiate, price and issue enterprise quotes.
category: Sales and Enrollment
trigger: A qualified enterprise opportunity needs a quote.
value: Faster, accurate quotes that shorten the sales cycle.
assumptions: Pricing is sourced from the master pricing system.

Candidate: Order Management
entity_id: VSR-0017
description: Capture and fulfil enterprise orders.
category: Fulfilment
trigger: An accepted quote becomes an order.
value: Reliable, on-time order fulfilment.
assumptions: Orders originate from approved quotes.

[... 48 more candidate blocks, all 50 governed Value Streams ...]
```

Output — `recommendations[]`:


```
{
  "recommendations": [
    { "valueStreamId": "VSR-0042", "valueStreamName": "Configure, Price and Quote", "confidence": 91,
      "supportType": "direct", "reason": "Ticket explicitly targets CPQ automation for enterprise quoting.", "sourceTickets": [] },
    { "valueStreamId": "VSR-0017", "valueStreamName": "Order Management", "confidence": 74,
      "supportType": "implied", "reason": "Quote-to-order handoff implied by the fulfilment requirement.", "sourceTickets": ["IDMT-16890"] }
  ]
}
```

Human approval gate — the SME confirms the Value Stream set before any generation runs.

---

3. Stage Selection (LLM ×1, all Value Streams batched)

Inputs:

- `content` → the ticket's raw text.
- `valueStreams` → each approved Value Stream with its **own candidate stages**; each stage block is `[sequence] Stage Name (stageId)` + description + entrance/exit criteria.

How the prompt tells the model to read stages: match the work's concrete **action** to a stage's **scope** — its description, entrance and exit criteria, value items, stakeholders — *not* on stage-name similarity; return only stage ids printed under that Value Stream; no count cap.

Filled prompt (user message):

```
## Ticket context
- content: {raw text}

## Approved value streams (each with its own candidate stages)
### Value Stream VSR-0042 — Configure, Price and Quote
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote.
entrance: opportunity qualified | exit: quote issued
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order.
entrance: quote accepted | exit: order created
[3] Quote Revision (VSS-0042-03)
description: Revise an issued quote on customer request.
entrance: revision requested | exit: revised quote issued

### Value Stream VSR-0017 — Order Management
[1] Order Capture (VSS-0017-01)
description: Receive and validate the enterprise order.
entrance: order requested | exit: order validated
[2] Order Fulfilment (VSS-0017-02)
description: Pick, pack and ship the order.
entrance: order confirmed | exit: order shipped
```

*The stage **name** is in each block (`[1] Opportunity to Quote (VSS-0042-01)`), so the model matches on scope and returns the stageId.*

Output — one entry per Value Stream:

```
{
  "valueStreams": [
    { "valueStreamId": "VSR-0042", "selectedStages": [
        { "stageId": "VSS-0042-01", "stageName": "Opportunity to Quote", "reason": "Ticket targets the quoting initiation workflow." },
        { "stageId": "VSS-0042-02", "stageName": "Quote to Order", "reason": "Quote-to-order handoff is in scope." } ] },
    { "valueStreamId": "VSR-0017", "selectedStages": [
        { "stageId": "VSS-0017-01", "stageName": "Order Capture", "reason": "Accepted quotes are captured as orders." } ] }
  ]
}
```

A stage placed under the wrong Value Stream is salvaged to its owner. Empty list = "take the whole governed list for the architect to trim."

---

4. Description BODY (LLM ×1, per ticket, VS-agnostic)

Inputs:

- `content` → the ticket's raw text.

System: write the shared narrative body; every statement must trace to a phrase in the content; availability/plan lines only when the content explicitly states them.

Filled prompt (user message):

```
Ticket context:
- content: {raw text}
```

Output — `{ text }`:

```
This theme automates enterprise quoting by integrating Salesforce CPQ with live Oracle ERP pricing,
cutting the quote cycle from five days to same-day. Discounts above 20% route to VP approval, and
accepted quotes hand off to order fulfilment within the Deal Desk's 4-hour SLA.
```

---

5. Description FRAMING (LLM ×1, all Value Streams batched)

Inputs:

- `content` → the ticket's raw text.
- `valueStreams` → each approved Value Stream: `valueStreamId, valueStreamName, valueStreamDescription, valueProposition`.

Filled prompt (user message):

```
Ticket context:
- content: {raw text}

Approved value streams:
- valueStreamId: VSR-0042
  valueStreamName: Configure, Price and Quote
  valueStreamDescription: Initiate, price and issue enterprise quotes.
  valueProposition: Faster, accurate quotes that shorten the sales cycle.
- valueStreamId: VSR-0017
  valueStreamName: Order Management
  valueStreamDescription: Capture and fulfil enterprise orders.
  valueProposition: Reliable, on-time order fulfilment.
```

Output — `framings[]`:

```
{
  "framings": [
    { "valueStreamId": "VSR-0042", "text": "Within Configure, Price and Quote, this initiative makes enterprise quoting real-time and rule-driven." },
    { "valueStreamId": "VSR-0017", "text": "For Order Management, it ensures accepted quotes flow cleanly into fulfilment." }
  ]
}
```

Final per-VS description = its framing paragraph + the shared body, concatenated deterministically.

---

6. Business Needs (LLM ×1 per Value Stream)

Inputs:

- `valueStream` → `valueStreamId, valueStreamName, valueStreamDescription, valueProposition`.
- `selectedStages` → that VS's selected stages (`[seq] Stage Name (stageId)` + description + criteria).
- `content` → the ticket's raw text.

System: write the consolidated Business Needs; every need, dependency, and rule must trace to a phrase in the content; one "Value Stage:" block per stage.

Filled prompt (user message):

```
## Approved value stream
ID: VSR-0042
Name: Configure, Price and Quote
Description: Initiate, price and issue enterprise quotes.
Value proposition: Faster, accurate quotes that shorten the sales cycle.

## Selected stages (write needs for these stages only)
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote. entrance: opportunity qualified | exit: quote issued
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order. entrance: quote accepted | exit: order created

## Ticket context
- content: {raw text}
```

Output — `{ text }` (one consolidated document):

```
Value Stage: Opportunity to Quote

Business Product Feature: Real-Time Quoting
1. Sales Operations requires real-time quote generation to replace the current 3-5 day manual cycle.
   Dependency: Salesforce CPQ and the Oracle pricing API.
   Business Rule: Discounts above 20% require VP approval.

Value Stage: Quote to Order

Business Product Feature: Order Handoff
1. Accepted quotes must hand off to order fulfilment within the Deal Desk 4-hour SLA.
```

---

7. Capabilities (L3) (LLM ×1, ALL Value Streams merged)

Inputs:

- `content` → the ticket's raw text.
- `valueStreams` → grouped **Value Stream** → **Stage** → **governed candidate L3**. Each candidate L3 line is `capabilityId | name - description [tier] (L2: parent)`.

System: for each stage, select L3 only from THAT stage's printed candidate list; strict stage isolation (the same id/name repeats across stages and Value Streams — match the exact printed id only). **L2 is derived 1-1 from the selected L3 — no separate L2 call.**

Filled prompt (user message):

```
## Ticket context
- content: {raw text}

## Approved value streams, each with its selected stages and each stage's own candidate L3
### Value Stream VSR-0042 — Configure, Price and Quote
description: Initiate, price and issue enterprise quotes.

### Stage VSS-0042-01
[1] Opportunity to Quote (VSS-0042-01)
description: Initiate and price an enterprise quote. entrance: opportunity qualified | exit: quote
issued
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0042-0011 | Real-Time Pricing Engine Integration - live CPQ pricing feed [tier: core] (L2:
Pricing and Discounting)
- L3-0042-0012 | Automated Discount Approval Workflow - routes discounts for approval (L2:
Approval and Governance)

### Stage VSS-0042-02
[2] Quote to Order (VSS-0042-02)
description: Convert an accepted quote into an order. entrance: quote accepted | exit: order
created
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0042-0021 | Order Handoff Orchestration - passes the quote to fulfilment (L2: Order
Management)

### Value Stream VSR-0017 — Order Management
description: Capture and fulfil enterprise orders.

### Stage VSS-0017-01
[1] Order Capture (VSS-0017-01)
description: Receive and validate the enterprise order. entrance: order requested | exit: order
validated
Candidate L3 capabilities (choose by id; each shows its parent L2):
- L3-0017-0011 | Order Intake Automation - captures order data (L2: Order Management)
```

Output — keyed by stageId (re-attributed to each Value Stream afterward):

```
{
  "stages": [
    { "stageId": "VSS-0042-01", "capabilities": [
        { "capabilityId": "L3-0042-0011", "capabilityName": "Real-Time Pricing Engine
Integration", "reason": "Live CPQ pricing feed from Oracle ERP." },
        { "capabilityId": "L3-0042-0012", "capabilityName": "Automated Discount Approval
Workflow", "reason": "Discounts above 20% require VP approval." } ] },
    { "stageId": "VSS-0042-02", "capabilities": [
        { "capabilityId": "L3-0042-0021", "capabilityName": "Order Handoff Orchestration",
"reason": "Accepted quotes hand off to fulfilment." } ] },
    { "stageId": "VSS-0017-01", "capabilities": [
        { "capabilityId": "L3-0017-0011", "capabilityName": "Order Intake Automation", "reason":
"Captures the order on quote acceptance." } ] }
  ]
}
```

*Strict isolation + salvage keep each L3 under its owning stage. **L2** = the unique parent L2s of the selected L3, derived per Value Stream.*

Final Theme package (deterministic — no LLM)

One package per approved Value Stream:

```
{
  "themeTitle": "Enabling Real-Time Quote Automation -- Configure, Price and Quote",
  "themeDescription": "<framing + body>",
  "selectedStages": [ { "stageId": "VSS-0042-01", "stageName": "Opportunity to Quote", "reason": "..." } ],
  "businessNeeds": "<consolidated text>",
  "l3Capabilities": [ { "stageId": "VSS-0042-01", "capabilities": [ ... ] } ],
  "l2Capabilities": [ { "stageId": "VSS-0042-01", "capabilities": [ { "capabilityId": "L2-0042-001", "name": "Pricing and Discounting" } ] } ]
}
```
